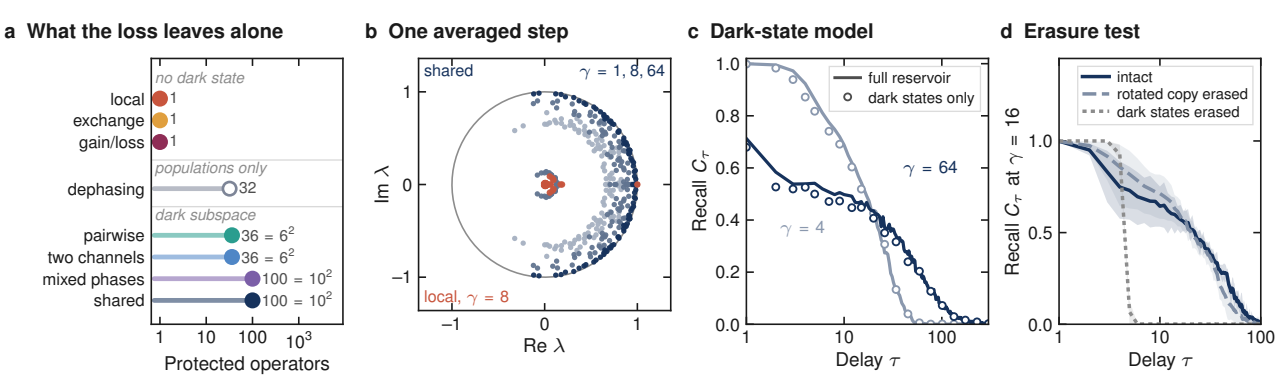

a What the loss leaves alone
no dark state
local
exchange
gain/loss
1
1
1
populations only
dephasing
32
dark subspace
pairwise
two channels
mixed phases
shared
36 = 6²
36 = 6²
100 = 10²
100 = 10²
1
10
100
10³
Protected operators
b One averaged step
shared
γ = 1, 8, 64
local, γ = 8
Im λ
Re λ
1
0
−1
−1
0
1
c Dark-state model
full reservoir
dark states only
γ = 64
γ = 4
Recall C_τ
Delay τ
1.0
0.8
0.6
0.4
0.2
0.0
1
10
100
d Erasure test
intact
rotated copy erased
dark states erased
Recall C_τ at $\gamma = 16$
Delay τ
1.0
0.5
0.0
1
10
100